Convergence of Independence Metropolis-Hastings Algorithms in High Dimensions: Iteration Lengths, Cutoff Profiles, and Diagnostic Unreliability

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Abstract

We characterize the high-dimensional convergence of the arithmetic mean from the independence Metropolis-Hastings algorithm initialized at its proposal. Convergence is measured in the worst-case expected absolute error over bounded functions. We prove necessary and sufficient conditions for the mean to converge in relation to the importance weight. We precisely describe the logarithmic cutoff profile of the sharp decay to convergence where the relative entropy between the target and proposal determines the required iteration length. These results make connections between the required iterations of independence Metropolis-Hastings and importance sampling. For concentrating Bayesian posteriors, we illustrate how the required iteration time can be estimated providing concrete guidance for simulations. We justify the logarithmic cutoff analysis by showing a standard cutoff phenomenon is absent and prove that popular convergence diagnostics based on simulation output can validate inaccurate simulations.

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1 Introduction

Metropolis-Hastings algorithms [Hastings, 1970, Tierney, 1998] are essential for generating samples to accurately estimate many high-dimensional integrals in statistics. The independence Metropolis-Hastings (IMH) algorithm is a simple version, yet still widely used as a final correction step for proposals generated from variational inference or generative artificial intelligence. Every proposal is drawn from a fixed distribution, independently of the current state, and a move from the previous state to the new state is accepted or rejected. Simulating the IMH algorithm does not require evaluating the normalizing constant of the target or proposal measures and unlike in general Metropolis-Hastings, the proposals and importance weights can be generated in parallel for the IMH algorithm. Theory developed for this algorithm is often a foundation for understanding more sophisticated Metropolis-Hastings algorithms.

This work is motivated by and draws strong parallels to the required number of samples for importance sampling in high dimensions [Chatterjee and Diaconis, 2018]. Unlike the requirements in importance sampling [Chatterjee and Diaconis, 2018, Agapiou et al., 2017, Sanz-Alonso, 2018], the required number of samples for an accurate simulation in general is not understood in high dimensions. There are many key differences between the IMH algorithm and importance sampling as the IMH algorithm simulates a dependent Markov chain and the arithmetic mean is used to estimate integrals instead of a weighted mean. The IMH algorithm has many interesting theoretical properties lacking in importance sampling such as convergence in total variation which is not available for importance sampling directly.

We characterize precisely the number of iterations for the high-dimensional L^1 convergence of the arithmetic mean of the IMH algorithm initialized at the proposal. The IMH algorithm must reduce the initialization bias and explore enough points for the arithmetic mean to reduce the L^1 error and approximate an integral well. We show an abrupt transition in the logarithmic number of iterations or a logarithmic cutoff phenomenon [Aldous and Diaconis, 1986, Chatterjee and Diaconis, 2018]. High dimensions here can mean a large state space dimension or large data set size. This is useful for studying high-dimensional probability measures or concentration of posteriors

for large sample sizes in Bayesian statistics.

Let $\mathbb{Z}_+$ denote the positive integers. Let $n \in \mathbb{Z}_+$ be an index parameter and let Π_n be a target probability measure on a Borel state space $\mathbf{X}_n$. Let $\mathcal{Q}_n$ be a proposal probability measure on $\mathbf{X}_n$ and $T_n \in \mathbb{Z}_+$ be an iteration time. Let $X_n(0) \sim \mathcal{Q}_n$ be initialized at the proposal, and $X_n(0), \ldots, X_n(T_n - 1)$ be samples from an independence Metropolis-Hastings (IMH) process. We study the precise convergence behavior of the L^1 loss as $n \to \infty$

$$R_{T_n}(\mathcal{Q}_n, \Pi_n) = \sup_{\|f\|_\infty \leq 1} \mathbb{E}\left(\left| \frac{1}{T_n} \sum_{t=0}^{T_n - 1} f(X_n(t)) - \int_{\mathbf{X}_n} f d\Pi_n \right| \right) \tag{1}$$

where the supremum is taken over Borel measurable functions $f : \mathbf{X}_n \to [-1, 1]$.

The contributions of this article are summarized as follows. Section 3 provides a complete high-dimensional convergence profile of the risk (1). Let w_n be an indexed importance weight and we prove that if $T_n \to \infty$, then $w_n/T_n \to 0$ in Π_n probability is necessary and sufficient for $R_{T_n}(\mathcal{Q}_n, \Pi_n) \to 0$. Section 3 also proves an abrupt transition towards convergence of (1) through a logarithmic cutoff and cutoff profile. Roughly, if the relative entropy $\mathbb{E}_{\Pi_n}(\log(w_n)) \to \infty$ and $\log(w_n)/\mathbb{E}_{\Pi_n}(\log(w_n)) \to 1$, we prove that $\log(T_n) \approx \mathbb{E}_{\Pi_n}(\log(w_n))$ is the logarithmic required number of iterations for (1) to be accurate. Section 4 illustrates the theory on concentrating Bayesian posteriors in Bayesian logistic regression with Laplace proposals where the required number of iterations can be estimated concretely. Section 5 proves that a standard cutoff to equilibrium in total variation is not possible for the IMH process. Section 6 provides further motivation for the theory by showing unreliability of convergence diagnostics such as the R-hat [Vehtari et al., 2021] for the IMH process. We prove convergence diagnostics can provide false positives and fail to be a reliable means of verifying convergence for the IMH algorithm.

The convergence behavior of the IMH algorithm has previously been studied often with bounded importance weights [Tierney, 1994]. Previous results address eigenvalues on finite state spaces [Liu, 1996] and hitting times [Maciuca and Zhu, 2006]. A quantitative characterization of geometric convergence from any initialization is known for

bounded importance weights [Mengersen and Tweedie, 1996]. The exact convergence from fixed point starts in both total variation and Wasserstein distance has been developed with bounded importance weights [Wang, 2022, Brown and Jones, 2024]. More recent results address unbounded importance weights [Deligiannidis et al., 2026] and [Daeijavad and Asoodeh, 2026]. Other results instead consider asymptotic-variance criteria [Deligiannidis and Lee, 2018] or study large deviations of the empirical measure [Milinanni and Nyquist, 2026]. There is related but different work on continuous-time Metropolis generators and high-dimensional independence-sampler jump limits [Choi and Huang, 2020, Lee and Neal, 2018].

2 Background and Motivation

We consider a sequence of target measures indexed by some parameter such as the dimension or data size. Let $(\mathbf{X}_n)_{n\geq 0}$ be a sequence of Borel measurable spaces equipped with their Borel sigma-algebra. For each $n \in \mathbb{Z}_+$, we are interested in estimating integrals from a Borel target probability measure Π_n using a proposal Borel probability measure $\mathcal{Q}_n$ on $\mathbf{X}_n$. We will assume we may choose an importance weight $w_n = d\Pi_n/d\mathcal{Q}_n$ that is everywhere finite on $\mathbf{X}_n$, which is possible when Π_n is absolutely continuous with respect to $\mathcal{Q}_n$.

Define the acceptance function $\alpha_n : \mathbf{X}_n \times \mathbf{X}_n \to [0,1]$ by

$$\alpha_n(x,y) = 1 \wedge \frac{w_n(y)}{w_n(x)}$$

if $w_n(x) > 0$ and 1 otherwise. Denote the Dirac measure at x by δ_x. Define the independence Metropolis-Hastings kernel for any $x \in \mathbf{X}_n$ and Borel $B \subseteq \mathbf{X}_n$ by

$$\mathcal{P}_n(x,B) = \int_B \alpha_n(x,y)\mathcal{Q}_n(dy) + r_n(x)\delta_x(B)$$

where the rejection probability is $r_n(x) = 1 - \int_{\mathbf{X}_n} \alpha_n(x,y)\mathcal{Q}_n(dy)$. We then define $\mathcal{P}_n^0(x,B) = \delta_x(B)$ and for $t \geq 1$, we define $\mathcal{P}_n^t(x,B) = \int \mathcal{P}_n(y,B)\mathcal{P}_n^{t-1}(x,dy)$.

The marginal distribution of $X_n(t)$ is denoted by $\mathcal{Q}_n\mathcal{P}_n^t$. With $X_n(0) \sim \mathcal{Q}_n$, this

Markov kernel defines an IMH process as a discrete-time Markov process $(X_n(t))_{t\geq 0}$. We will only consider the proposal start process throughout the manuscript. We may view $\mathcal{Q}_n$ as the best approximation available to Π_n and this is then the optimal initialization available in practice and the initialization we consider.

3 High-dimensional Convergence Theory and Cut-off Profiles

Define the ϵ-approximation time of a risk for any $\epsilon \in (0, 2)$

$$T_\epsilon(\mathcal{Q}_n, \Pi_n) = \inf\{T \in \mathbb{Z}_+ : R_T(\mathcal{Q}_n, \Pi_n) \leq \epsilon\}$$

and $\inf \varnothing = \infty$. Our goal is to characterize the sharp decrease in the risk (1) visualized in Figure 1. The goal here is to characterize the required number of samples for Metropolis-Hastings in high dimensions. This is a sharp phase transition called cutoff.

We have the following sharp characterization of high-dimensional convergence in terms of the importance weight.

Theorem 1. *Suppose the iteration length $T_n \to \infty$. Then,*

$$\lim_{n\to\infty} \Pi_n \left(\frac{w_n}{T_n} \leq \delta \right) = 1$$

for every $\delta \in (0, 1)$ is equivalent to

$$R_{T_n}(\mathcal{Q}_n, \Pi_n) \to 0.$$

We can characterize more precisely the sharp decrease of the risk in terms of the relative entropy. The main investigation is how the log weight concentrates towards the relative entropy $\mathcal{H}(\Pi_n, \mathcal{Q}_n) = \int \log(w_n) d\Pi_n$ so that in high dimensions,

$$\log(w_n) \approx \mathcal{H}(\Pi_n, \mathcal{Q}_n).$$

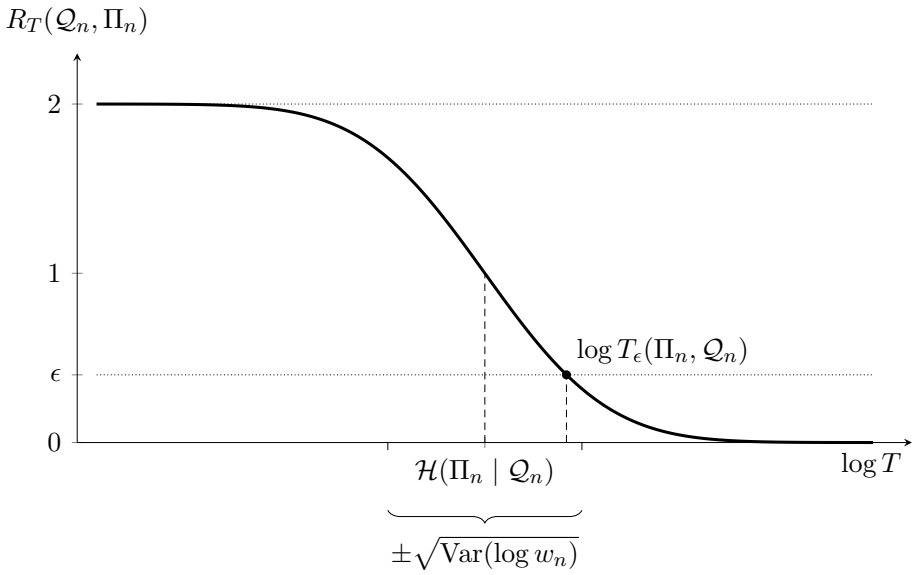


Figure 1: A visualization of the cutoff for the IMH process. As $\log T$ passes the relative entropy $\mathcal{H}(\Pi_n, \mathcal{Q}_n)$, the risk $R_T(\mathcal{Q}_n, \Pi_n)$ falls from 2 to 0 along the profile $2\{1 - \Phi(z)\}$ of Theorem 2.

Let $N(0,1)$ denote the standard Gaussian distribution and Φ be the distribution function of a standard Gaussian random variable. We will characterize the logarithmic cutoff profile through the asymptotic normality of

$$Z_n = \frac{\log(w_n) - \mathcal{H}(\Pi_n, \mathcal{Q}_n)}{\sqrt{\mathrm{Var}_{\Pi_n}(\log(w_n))}}. \tag{2}$$

Theorem 2. *For $n \in \mathbb{Z}_+$, assume $\mathcal{H}(\Pi_n, \mathcal{Q}_n) < \infty$ and $\lim_{n\to\infty} \mathcal{H}(\Pi_n, \mathcal{Q}_n)) = \infty$. If for any $\delta \in (0,2)$,*

$$\lim_{n\to\infty} \Pi_n \left(\left| \frac{\log(w_n)}{\mathcal{H}(\Pi_n, \mathcal{Q}_n)} - 1 \right| \le \delta \right) = 1,$$

then for any $\epsilon \in (0,2)$,

$$\lim_{n\to\infty} \frac{\log(T_\epsilon(\mathcal{Q}_n, \Pi_n))}{\mathcal{H}(\Pi_n, \mathcal{Q}_n)} = 1. \tag{3}$$

Assume $0 < Var_{\Pi_n}(\log(w_n)) < \infty$ *and* $Var_{\Pi_n}(\log(w_n)) \to \infty$ *and*

$$Z_n \to N(0,1)$$

weakly in Π_n*. For* $z \in \mathbb{R}$*, let* $T_n(z) = 1 \vee \lfloor \exp(\mathcal{H}(\Pi_n, \mathcal{Q}_n) + z\sqrt{Var_{\Pi_n}(\log(w_n))}) \rfloor$ *be an iteration time. Then*

$$\lim_{n\to\infty} R_{T_n(z)}(\mathcal{Q}_n, \Pi_n) = 2(1 - \Phi(z)). \tag{4}$$

The second conclusion describes the phase transition in terms of the variance of the log weight or the varentropy which has more recently shown up in Markov process cutoff theory [Salez, 2026]. We develop the proofs of Theorem 1 and Theorem 2. We first prove a general upper and lower bound on the risk.

Theorem 3. *For any Borel probability measures* $\Pi, \mathcal{Q}$ *on a Borel space* $\boldsymbol{X}$ *with importance weight* $w = d\Pi/d\mathcal{Q}$ *that is everywhere finite on* $\boldsymbol{X}$*, and for all* $T \geq 1$*,*

$$R_T(\mathcal{Q}, \Pi) \geq 2 \sup_{B \subseteq \boldsymbol{X}, B\, Borel} \left\{ \Pi(B) \left(1 - Q(B)\right)^T \right\}.$$

Let $u > 0$ *and* $C = \{w \leq u\}$ *such that* $\Pi(C) > 0$ *and let* $\lambda_C = 1 - \mathcal{Q}(C) + \Pi(C)/u$*. Then*

$$R_T(\mathcal{Q}, \Pi) \leq \sqrt{\frac{2}{T\lambda_C}} + \frac{2\mathcal{Q}(C)}{T\lambda_C} + 2\int_{C^c} T \wedge \frac{w}{\Pi(C)} d\mathcal{Q} + 2(1 - \Pi(C)).$$

Proof of Theorem 3. The problem that yields the lower bound is a tail distribution mismatch. Let $B \subseteq \mathbf{X}$ be Borel. Then we have

$$\sup_{x \in B^c} \mathcal{P}(x, B) \leq Q(B)$$

and therefore,

$$\inf_{x \in B^c} \mathcal{P}(x, B^c) \geq 1 - Q(B).$$

Let $B_T^c = \{X_k \in B^c, 0 \leq k \leq T-1\}$ and let $\varphi = I_B - I_{B^c}$. Then on B_T^c,

$$\frac{1}{T}\sum_{k=0}^{T-1}\varphi(X_k) - \int_{\mathbf{X}}\varphi d\Pi = -2\Pi(B).$$

This gives the lower bound

$$\begin{aligned}\mathbb{E}\left(\left|\frac{1}{T}\sum_{k=0}^{T-1}\varphi(X_k) - \int_{\mathbf{X}}\varphi d\Pi\right|\right) &\geq \mathbb{E}\left(\left|\frac{1}{T}\sum_{k=0}^{T-1}\varphi(X_k) - \int_{\mathbf{X}}\varphi d\Pi\right| I_{B_T^c}\right)\\ &\geq 2\Pi(B)\mathbb{E}\left(I_{B_T^c}\right)\\ &\geq 2\Pi(B)\left(1-Q(B)\right)^T.\end{aligned}$$

We now prove the upper bound. The technique here is essentially that of a drift and minorization argument. In particular, IMH samples $\Pi_C = \Pi(\cdot \mid C)$ and this can be controlled by a standard convergence argument conditioned on this set and Lyapunov control off of this set. Let $\varphi : \mathbf{X} \to \mathbb{R}$ with $\|\varphi\|_\infty \leq 1$ and let $\psi = \varphi - \int_{\mathbf{X}}\varphi(x)\Pi_C(dx)$. This gives the upper bound through the decomposition

$$\begin{aligned}&\left|\frac{1}{T}\sum_{k=0}^{T-1}\varphi(X_k) - \int_{\mathbf{X}}\varphi d\Pi\right|\\ &\leq \left|\frac{1}{T}\sum_{k=0}^{T-1} I_C(X_k)\psi(X_k)\right| + \left|\frac{1}{T}\sum_{k=0}^{T-1}\psi(X_k)I_{C^c}(X_k)\right| + \left|\int_{\mathbf{X}}\psi d\Pi\right|.\end{aligned}$$

The last term is controlled by

$$\left|\int_{\mathbf{X}}\varphi(x)\Pi_C(dx) - \int_{\mathbf{X}}\varphi d\Pi\right| \leq 2\Pi(C^c) = 2(1-\Pi(C)).$$

The first term is bounded by a fairly standard convergence argument with bias and variance control. Let $\mathcal{P}_C$ denote IMH with proposal $\mathcal{Q}_C = \mathcal{Q}(\cdot \mid C)$ and target Π_C. This restricted Markov kernel satisfies a standard global minorization. For any Borel

$B \subseteq \mathbf{X}$ and $x \in C$,

$$\mathcal{P}_C(x, B) \geq \int_{B \cap C} \frac{\alpha(x, y)}{\mathcal{Q}(C)} \mathcal{Q}(dy) \geq \frac{1}{\mathcal{Q}(C)} \int_{B \cap C} \frac{w(y)}{u} \mathcal{Q}(dy) = \frac{\Pi(C)}{\mathcal{Q}(C)u} \Pi_C(B).$$

Changing the measure we can see $\frac{\Pi(C)}{\mathcal{Q}(C)u} \leq 1$. We can write the restricted residual kernel for $x \in C$, if $\frac{\Pi(C)}{\mathcal{Q}(C)u} < 1$

$$R_C(x, \cdot) = \frac{\mathcal{P}_C(x, \cdot) - \frac{\Pi(C)}{\mathcal{Q}(C)u} \Pi_C(\cdot)}{1 - \frac{\Pi(C)}{\mathcal{Q}(C)u}}$$

and if $\frac{\Pi(C)}{\mathcal{Q}(C)u} = 1$, $R_C(x, \cdot) = \Pi_C$. Let $\lambda_C = 1 - \mathcal{Q}(C) + \frac{\Pi(C)}{u}$. Let

$$\nu(\cdot) = \left[\mathcal{Q}(\cdot \cap C^c) + \Pi(C)\Pi_C(\cdot)/u\right] / \lambda_C.$$

On C^c, the proposal is accepted and so we may write for $x \in C$,

$$\begin{aligned}
\mathcal{P}(x, \cdot) &= \mathcal{Q}(\cdot \cap C^c) + \mathcal{Q}(C)\mathcal{P}_C(x, \cdot) \\
&= \mathcal{Q}(\cdot \cap C^c) + \frac{\Pi(C)}{u} \Pi_C(\cdot) + \left(\mathcal{Q}(C) - \frac{\Pi(C)}{u}\right) R_C(x, \cdot) \\
&= \lambda_C \nu(\cdot) + (1 - \lambda_C) R_C(x, \cdot).
\end{aligned}$$

So then

$$\sup_{x, y \in C} \|\mathcal{P}(x, \cdot) - \mathcal{P}(y, \cdot)\|_{TV} \leq 1 - \lambda_C.$$

Let $\mu(\cdot) = \mathcal{Q}(\cdot \cap C^c) + \mathcal{Q}(C)\Pi_C(\cdot)$ and define the IMH process $(Y_t)_{t \geq 0}$ with $Y_0 \sim \mu$. Let $S_T(\psi) = \sum_{k=0}^{T-1} I_C(X_k)\psi(X_k)$ and let $S'_T(\psi) = \sum_{k=0}^{T-1} I_C(Y_k)\psi(Y_k)$. So then

$$\|\mathcal{Q} - \mu\|_{TV} \leq \mathcal{Q}(C).$$

We can control the bias with a coupling convergence argument. On C^c, X_0, Y_0 are the same with probability $\mathcal{Q}(C^c)$ and on entering C, the split chain gives the rate to couple

$1 - \lambda_C$. The upper bound is

$$\mathbb{E}\left|\frac{1}{T}S_T(\psi) - \frac{1}{T}S'_T(\psi)\right| \leq \frac{2\mathcal{Q}(C)}{T}\sum_{k=0}^{T-1}(1-\lambda_C)^k \leq \frac{2\mathcal{Q}(C)}{T\lambda_C}.$$

We can also control the variance. Define $\psi_C = \psi I_C$. Then $\mathcal{P}^t\psi_C = I_C(x)\mathcal{Q}(C)^t\mathcal{P}_C^t\psi$ for $t \in \mathbb{Z}_+$. For $i \leq j$,

$$\mathbb{E}(\psi_C(Y_i)\psi_C(Y_j)) \leq \left|\mathcal{Q}(C)^{j-i}\Pi_C(\psi_C\mathcal{P}_C^{j-i}\psi_C)\right| \leq (1-\lambda_C)^{j-i}.$$

Then

$$\mathbb{E}\left|\frac{1}{T}S'_T(\psi)\right|^2 \leq \frac{1}{T} + \frac{2}{T}\sum_{k=1}^{T-1}\left(1-\frac{k}{T}\right)(1-\lambda_C)^k \leq \frac{2}{T\lambda_C}.$$

The second term is controlled through an interesting reverse Lyapunov drift to C on the density function. For $k \in \mathbb{Z}_+$, let $D_k = \frac{d\mathcal{Q}\mathcal{P}^k}{d\mathcal{Q}}$ and $D_0 = 1$. Then

$$\begin{aligned}\int_B D_{k+1}d\mathcal{Q} &= \mathcal{Q}\mathcal{P}^{k+1}(B) = \mathcal{Q}\mathcal{P}^k\mathcal{P}(B)\\ &= \int_B\int_{\mathbf{X}} D_k(x)\alpha(x,y)\mathcal{Q}(dx)\mathcal{Q}(dy) + \int_B r(x)D_k(x)\mathcal{Q}(dx).\end{aligned}$$

Therefore for $\mathcal{Q}$ almost every $x \in \mathbf{X}$,

$$D_{k+1}(x) = \int_{\mathbf{X}} D_k(y)\alpha(y,x)\mathcal{Q}(dy) + r(x)D_k(x).$$

For $x \notin C$ and $y \in C$, $w(y) \leq u < w(x)$ and $w(y)/w(x) \leq 1$, and the acceptance is bounded below by

$$\int_{\mathbf{X}} \alpha(x,y)\mathcal{Q}(dy) \geq \int_C \frac{w(y)}{w(x)}\mathcal{Q}(dy) = \frac{\Pi(C)}{w(x)}.$$

By construction $\int_{\mathbf{X}} D_k(y)\mathcal{Q}(dy) = 1$. We have by recursion for $x \notin C$, $D_0 = 1$ and for

$k \geq 1$,

$$D_k(x) \leq 1 + \left(1 - \frac{\Pi(C)}{w(x)}\right) D_{k-1}(x) \leq \sum_{i=0}^{k} \left(1 - \frac{\Pi(C)}{w(x)}\right)^i.$$

By Bernoulli's inequality,

$$\sum_{i=0}^{k} \left(1 - \frac{\Pi(C)}{w(x)}\right)^i \leq \frac{1 - \left(1 - \frac{\Pi(C)}{w(x)}\right)^{k+1}}{\frac{\Pi(C)}{w(x)}} \leq (k+1) \wedge \left(\frac{w(x)}{\Pi(C)}\right).$$

Integrating over the initial condition for every $T \geq 1$,

$$\mathbb{E}\left[\frac{1}{T}\sum_{k=0}^{T-1} I_{C^c}(X_k)\right] \leq \frac{1}{T}\sum_{k=0}^{T-1} \int_{C^c} (k+1) \wedge \left(\frac{w(x)}{\Pi(C)}\right) \mathcal{Q}(dx) \leq \int_{C^c} T \wedge \left(\frac{w(x)}{\Pi(C)}\right) \mathcal{Q}(dx).$$

□

This result allows us to prove a strong result for sequences of probability measures, but the result is less descriptive.

Lemma 4. *Let $T_n \to \infty$ and assume there is an $L_n \to \infty$ such that*

$$\lim_{n\to\infty} \Pi_n\left(\frac{w_n}{T_n} \leq \frac{1}{L_n}\right) = h \in (0,1] \qquad \lim_{n\to\infty} \Pi_n\left(\frac{1}{L_n} \leq \frac{w_n}{T_n} \leq L_n\right) = 0. \tag{5}$$

Then

$$\lim_{n\to\infty} R_{T_n}(\mathcal{Q}_n, \Pi_n) = 2(1-h).$$

Proof. Assume $n \in \mathbb{Z}_+$ is large enough so $T_n L_n > 1$. Define the set $C_n = \{w_n/T_n \leq L_n\}$. For the lower bound, using the lower bound in Theorem 3 with Markov's inequal-

ity and Bernoulli's inequality,

$$\begin{aligned} R_{T_n}(\mathcal{Q}_n, \Pi_n) &\geq 2\left[1 - \Pi_n\left(C_n\right)\right]\left(1 - \mathcal{Q}_n\left(C_n^c\right)\right)^{T_n} \\ &\geq 2\left[1 - \Pi_n\left(C_n\right)\right]\left(1 - \frac{1}{L_n T_n}\right)^{T_n} \\ &\geq 2\left[1 - \Pi_n\left(C_n\right)\right]\left(1 - 1/L_n\right). \end{aligned}$$

Using the assumed conditions (5), as $n \to \infty$,

$$\Pi_n\left(C_n^c\right) = 1 - \Pi_n\left(\frac{w_n}{T_n} \leq 1/L_n\right) - \Pi_n\left(1/L_n < \frac{w_n}{T_n} \leq L_n\right) \to 1 - h.$$

Therefore,

$$\liminf_{n\to\infty} R_{T_n}(\mathcal{Q}_n, \Pi_n) \geq 2(1-h).$$

Define the set $D_n = \{w_n/T_n \leq 1/L_n\}$. Let $\lambda_n = 1 - \mathcal{Q}_n(D_n) + L_n \Pi_n(D_n)/T_n$. For the upper bound, we use Theorem 3,

$$R_{T_n}(\mathcal{Q}_n, \Pi_n) \leq \sqrt{\frac{2}{T_n \lambda_n}} + \frac{2\mathcal{Q}_n(D_n)}{T_n \lambda_n} + 2\int_{D_n^c} T_n \wedge \frac{w}{\Pi(D_n)} d\mathcal{Q}_n + 2\Pi(D_n^c).$$

Then $\lambda_n T_n \to \infty$ and the first two terms tend to 0 as $n \to \infty$. By (5), the last term $2\Pi(D_n^c) \to 2(1-h)$. It remains to show the remaining term vanishes and we have

$$\int_{\left\{\frac{w_n}{T_n} > \frac{1}{L_n}\right\}} T_n \wedge \frac{w_n}{\Pi_n(D_n)} d\mathcal{Q}_n = \int_{\left\{\frac{w_n}{T_n} > \frac{1}{L_n}\right\}} \frac{T_n}{w_n} \wedge \frac{1}{\Pi_n(D_n)} d\Pi_n$$

and

$$\begin{aligned} \int_{\left\{\frac{w_n}{T_n} > \frac{1}{L_n}\right\}} \frac{T_n}{w_n(x)} \wedge \frac{1}{\Pi_n(D_n)} \Pi_n(dx) &\leq \int_{\left\{\frac{1}{L_n} < \frac{w_n}{T_n} \leq L_n\right\}} \frac{T_n}{w_n(x)} \wedge \frac{1}{\Pi_n(D_n)} \Pi_n(dx) \\ &\quad + \int_{\left\{L_n < \frac{w_n}{T_n}\right\}} \frac{T_n}{w_n(x)} \wedge \frac{1}{\Pi_n(D_n)} \Pi_n(dx) \\ &\leq \frac{1}{\Pi_n(D_n)} \Pi_n\left(\frac{1}{L_n} < \frac{w_n}{T_n} \leq L_n\right) + \frac{1}{L_n}. \end{aligned}$$

By the assumed conditions (5), both terms tend to 0 as $n \to \infty$. Therefore,

$$\limsup_{n\to\infty} R_{T_n}(\mathcal{Q}_n, \Pi_n) \le 2(1-h).$$

□

We are now in a position to prove Theorem 1 and Theorem 2.

Proof of Theorem 1. Assume $w_n/T_n \not\to 0$ in probability. Then there are $\epsilon, \delta \in (0,1)$ and a subsequence satisfying $\Pi_{n_k}(w_{n_k}/T_{n_k} \ge \epsilon) \ge \delta$. Using the lower bound in Theorem 3 and Markov's inequality, then for all sufficiently large n,

$$\begin{aligned} R_{T_{n_k}}(\mathcal{Q}_{n_k}, \Pi_{n_k}) &\ge 2\Pi_{n_k}(w_{n_k}/T_{n_k} \ge \epsilon)\left(1 - \frac{1}{\epsilon T_{n_k}}\right)^{T_{n_k}} \\ &\ge 2\delta\left(1 - \frac{1}{\epsilon T_{n_k}}\right)^{T_{n_k}} \\ &\to 2\delta \exp(-1/\epsilon). \end{aligned}$$

Therefore, $R_{T_n}(\mathcal{Q}_n, \Pi_n) \not\to 0$.

Now assume $w_n/T_n \to 0$ in probability. Let $C_n = \{w_n \le T_n\epsilon\}$. Then for any $\epsilon > 0$, $\Pi_n(C_n^c) \to 0$. Let $\lambda_{C_n} = 1 - \mathcal{Q}_n(C_n) + \Pi_n(C_n)/(\epsilon T_n)$. The upper bound in Theorem 3 is

$$\begin{aligned} R_{T_n}(\mathcal{Q}_n, \Pi_n) &\le \sqrt{\frac{2}{T_n\lambda_{C_n}}} + \frac{2\mathcal{Q}(C_n)}{T_n\lambda_{C_n}} + 2\int_{C_n^c} T_n \wedge \frac{w_n}{\Pi(C_n)} d\mathcal{Q}_n + 2\Pi(C_n^c) \\ &\le \sqrt{\frac{2\epsilon}{\Pi(C_n)}} + \frac{2\epsilon\mathcal{Q}(C_n)}{\Pi(C_n)} + 2\frac{\Pi(C_n^c)}{\Pi(C_n)} + 2\Pi(C_n^c). \end{aligned}$$

Since $\Pi(C_n) \to 1$ and $\epsilon > 0$ is arbitrary, then $R_{T_n} \to 0$. □

Proof of Theorem 2. We first prove the logarithmic cutoff (3). Let $\epsilon, \delta > 0$ and define

$$M_n^+(\delta) = \lfloor \exp[(1+\delta)\mathcal{H}(\Pi_n, \mathcal{Q}_n)] \rfloor, \qquad M_n^-(\delta) = \lfloor \exp[(1-\delta)\mathcal{H}(\Pi_n, \mathcal{Q}_n)] \rfloor.$$

Let $0 < s < \delta$ and choose $L_n = \exp(s\mathcal{H}(\Pi_n, \mathcal{Q}_n))$. Then

$$\Pi_n(\log(w_n) - \log(M_n^+(\delta)) \leq -\log(L_n)) \to 1.$$

Therefore Lemma 4 yields $T_\epsilon(\mathcal{Q}_n, \Pi_n) \leq M_n^+(\delta)$ for sufficiently large n. Now for the lower bound. Let $B_n = \{w_n \geq \exp[(1-\delta/2)\mathcal{H}(\Pi_n, \mathcal{Q}_n)]\}$ so that by the consistency assumption, $\Pi_n(B_n) \to 1$. With Markov's inequality and Theorem 3, for any $T \leq M_n^-(\delta)$

$$\begin{aligned} R_T(\mathcal{Q}_n, \Pi_n) &\geq 2\Pi_n(B_n)(1 - \mathcal{Q}_n(B_n))^T \\ &\geq 2\Pi_n(B_n)(1 - T\mathcal{Q}_n(B_n)) \\ &\geq 2\Pi_n(B_n)\left\{1 - T\exp\left[-(1-\delta/2)\mathcal{H}(\Pi_n, \mathcal{Q}_n)\right]\right\} \\ &\to 2. \end{aligned}$$

It follows that for sufficiently large n

$$M_n^-(\delta) \leq T_\epsilon(\mathcal{Q}_n, \Pi_n) \leq M_n^+(\delta).$$

Therefore taking $\delta \to 0$, then we obtain the conclusion.

We now prove the cutoff profile (4). The Gaussian limit assumption implies

$$\mathbb{E}_{\Pi_n}(\log(w_n))/\mathrm{Var}_{\Pi_n}(\log(w_n)]^{1/2} \to \infty$$

or the Gaussian limit would fail. For $z \in \mathbb{R}$, this implies $T_n(z) \to \infty$. Choose $L_n = \exp([\mathrm{Var}_{\Pi_n}(\log(w_n)])^{1/4})$ and by assumption $L_n \to \infty$. We will show the conditions of Lemma 4 hold. By the assumed condition,

$$\begin{aligned} &\Pi_n\left(w_n/T_n(z) \leq 1/L_n\right) \\ &= \Pi_n\left(\frac{\log(w_n) - \mathbb{E}_{\Pi_n}(\log(w_n))}{\sqrt{\mathrm{Var}(\log(w_n))}} \leq \frac{\log(T_n(z)/L_n) - \mathbb{E}_{\Pi_n}(\log(w_n))}{\sqrt{\mathrm{Var}(\log(w_n))}}\right). \end{aligned}$$

Now,

$$\frac{\log(T_n(z)/L_n) - \mathbb{E}_{\Pi_n}(\log(w_n))}{\sqrt{\mathrm{Var}(\log(w_n))}} \to z.$$

Since Φ is continuous, then $\Pi_n\left(w_n/T_n(z) \leq 1/L_n\right) \to \Phi(z)$. Similarly,

$$\frac{\log(T_n(z)L_n) - \mathbb{E}_{\Pi_n}(\log(w_n))}{\sqrt{\mathrm{Var}(\log(w_n))}} \to z$$

and so $\Pi_n\left(w_n/T_n(z) \leq L_n\right) \to \Phi(z)$. Therefore, $\Pi_n\left(1/L_n \leq w_n/T_n(z) \leq L_n\right) \to 0$. □

4 Applications

We consider some informative examples that highlight the theory developed. The Gaussian mismatch examples illustrate proposal mismatches to the target which are often exhibited in practice. The Bayesian concentration example highlights the computability of the logarithmic cutoff profile.

4.1 Application: Gaussian mean mismatches

For $d \in \mathbb{Z}_+$, let $\mathbf{X}_d = \mathbb{R}^d$ and consider $\Pi_d \equiv N(\mu_d, I_d)$ and $\mathcal{Q}_d \equiv N(0, I_d)$ where $\|\mu_d\| \to \infty$. A standard computation yields

$$\log(w_d(x)) = \mu_d^T x - \frac{1}{2}\|\mu_d\|^2.$$

Then $\mathcal{H}(\Pi_d, \mathcal{Q}_d) = \frac{1}{2}\|\mu_d\|^2$ and $\mathrm{Var}_{\Pi_d}[\log(w_d(x))] = \|\mu_d\|^2$, so

$$\frac{\log(w_d(x)) - \frac{1}{2}\|\mu_d\|^2}{\|\mu_d\|}$$

is a standard $N(0,1)$ random variable. Theorem 2 yields the cutoff profile

$$\lim_{d\to\infty} R_{T_d(z)}(\mathcal{Q}_d, \Pi_d) = 2(1 - \Phi(z))$$

where for $z \in \mathbb{R}$,

$$T_d(z) = 1 \vee \left\lfloor \exp\left(\frac{1}{2}\|\mu_d\|^2 + z\|\mu_d\|\right)\right\rfloor.$$

In particular, choose $\mu_d \equiv (1, \ldots, 1)^T \in \mathbb{R}^d$ and the number of samples required can be exponential in the dimension since $T_d(z) = 1 \vee \left\lfloor \exp\left(\frac{d}{2} + z\sqrt{d}\right)\right\rfloor$.

4.2 Application: Gaussian variance mismatches

Let $\mathbf{X} = \mathbb{R}^d$ and consider $\Pi_d \equiv N(0, I_d)$ and $\mathcal{Q}_d \equiv N(0, \tau^2 I_d)$ where $\tau > 0$ and $\tau \neq 1$. We have

$$\log(w_d(x)) = d\log(\tau) + \frac{1/\tau^2 - 1}{2}\|x\|^2.$$

Under Π_d, $\|X\|^2$ has mean d and variance $2d$. Then

$$\frac{\log(w_d) - \mathbb{E}_{\Pi_d}[\log(w_d)]}{\sqrt{\mathrm{Var}_{\Pi_d}[\log(w_d)]}} = \mathrm{sgn}(1/\tau^2 - 1)\left[\frac{\|X\|^2 - d}{\sqrt{2d}}\right] \to N(0,1)$$

weakly as $d \to \infty$. For $z \in \mathbb{R}$, let

$$T_d(z) = 1 \vee \left\lfloor \exp\left(d\log(\tau) + \frac{d(1/\tau^2 - 1)}{2} + z\sqrt{\frac{d}{2}}|1/\tau^2 - 1|\right)\right\rfloor$$

and with Φ the distribution function of a Gaussian, Theorem 2 yields the cutoff profile

$$\lim_{d\to\infty} R_{T_d(z)}(\mathcal{Q}_d, \Pi_d) = 2(1 - \Phi(z)).$$

In particular, the number of samples required is exponential in the dimension.

4.3 Application: Bayesian Posterior Concentration

Consider Bayesian logistic regression with a Gaussian prior and the IMH algorithm with an independent Laplace proposal. Let $N \in \mathbb{Z}_+$ denote the size of the data and let $d \equiv d_N \in \mathbb{Z}_+$ denote the dimension of the features. We will use the logarithmic cutoff

profile to determine how many IMH iterations are needed to correct the approximation when computing posterior integrals. Let $\tau \in (0, \infty)$ be a fixed variance parameter and we will assume a Gaussian prior $\beta \sim N(0, \tau^2 I_d)$. Let $Y_i \in \{0, 1\}$ and $x_i \in \mathbb{R}^d$ for $i \in 1, \ldots, N$ and the posterior $\Pi_N \propto \exp(-U_N(\beta))d\beta$ has potential

$$U_N(\beta) = \sum_{i=1}^{N} \left[\log(1 + \exp(x_i^T \beta)) - Y_i x_i^T \beta\right] + \frac{1}{2\tau^2} \|\beta\|^2 . \tag{6}$$

Let $\widehat{\beta}_N$ denote its unique minimal point, which exists. Let $\widehat{H}_N = \nabla^2 U_N(\widehat{\beta}_N)$ and define the Laplace proposal $\mathcal{Q}_N \equiv N(\widehat{\beta}_N, \widehat{H}_N^{-1})$.

We will consider the following assumptions on the data used in the posterior. Assume the design rows $x_1, \ldots, x_N$ are independent and have the form

$$x_i = (1, Z_1^i, \ldots, Z_{d-1}^i)^T, \qquad \mathbb{E}(Z_j^i) = 0, \quad \mathbb{E}(Z_j^{i\,2}) = 1, \quad |Z_j^i| \leq K$$

for some fixed $K \in (0, \infty)$ and for each i, the coordinates $Z_1^i, \ldots, Z_{d-1}^i$ are independent. Let $\sigma(t) = 1/(1 + e^{-t})$. Assume for some fixed $B \in (0, \infty)$ and β_d^* with $\|\beta_d^*\| \leq B$, the responses are conditionally independent with

$$Y_i \mid (x_j)_{j=1}^N \sim \text{Bernoulli}\big(\sigma(x_i^T \beta_d^*)\big). \tag{7}$$

Write $\mathbb{P}_*$ and $\mathbb{E}_*$ for sampling probability and expectation for $(x_i, Y_i)_{i=1}^N$. The following constructs a cutoff profile that can be computed from the data in a simulation where the risk (1) studied here is conditioned on the data.

Theorem 5. *Let $(x_i, Y_i)_{i=1}^N$ be samples from $\mathbb{P}_*$. Conditioned on the data, let Π_N denote the Bayesian logistic posterior and $\mathcal{Q}_N$ the Laplace proposal defined in this section. Assume $d^{5/3}/N \to 0$. Define the fitted shift vector*

$$v_N = \frac{1}{2} \widehat{H}_N^{-1/2} \sum_{i=1}^{N} \sigma''(x_i^T \widehat{\beta}_N)(x_i^T \widehat{H}_N^{-1} x_i) x_i \tag{8}$$

and assume $\|v_N\| \to \infty$ in $\mathbb{P}_*$ probability. For every $z \in \mathbb{R}$, with $T_N(z) = 1 \vee$

$\lfloor \exp\left(\|v_N\|^2/2 + z\|v_N\|\right)\rfloor$,

$$R_{T_N(z)}(\mathcal{Q}_N, \Pi_N) \to 2(1-\Phi(z))$$

in $\mathbb{P}_*$ *probability.*

5 Absence of Standard Cutoff

We motivate the logarithmic cutoff theory in the L^1 risk (1) and prove that there is no total variation cutoff for the marginal distribution of the IMH process. Define the total variation for two Borel probability measures ν, ν' on $\mathbf{X}$

$$\|\nu - \nu'\|_{TV} = \sup_{f:\mathbf{X}\to[0,1]} \left| \int f d\nu - \int f d\nu' \right|$$

where the functions in the supremum are Borel measurable. For $n \in \mathbb{Z}_+$ and $\epsilon \in (0,1)$, define

$$\tau_n(\epsilon) = \inf\{t \in \mathbb{Z}_+ \cup \{0\} : \left\|\mathcal{Q}_n \mathcal{P}_n^t - \Pi_n\right\|_{TV} \le \epsilon\}$$

where $\inf \varnothing = \infty$. Convergence of the IMH process exhibits a sharp transition to convergence or cutoff in total variation on a standard time scale if for $\epsilon \in (0, 1/2)$, $\tau_n(1-\epsilon) \to \infty$ and

$$\lim_{n\to\infty} \frac{\tau_n(\epsilon)}{\tau_n(1-\epsilon)} = 1.$$

The following shows this is not possible. Figure 2 provides a visualization.

Theorem 6. *For any* $\epsilon \in (0, 1/2)$*, if* $\tau_n(1-\epsilon) \to \infty$*, then*

$$\liminf_{n\to\infty} \frac{\tau_n(\epsilon)}{\tau_n(1-\epsilon)} > 1.$$

Proof. We decompose $\mathcal{Q}\mathcal{P}^k$ into its singular part with respect to Π and the residual part using ideas from [Smith and Tierney, 1996] to get a multiplicative decomposition

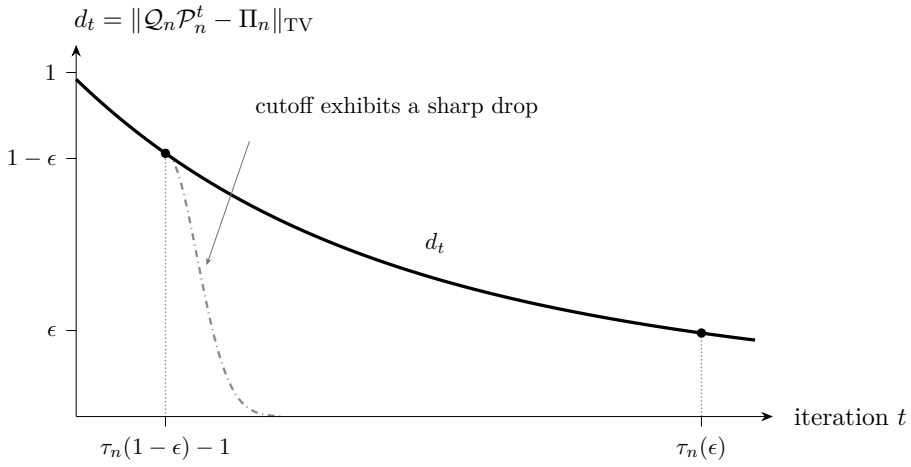


Figure 2: A visualization of the main proof idea for Theorem 6 showing the decay of total variation from time $\tau_n(1-\epsilon)$ to $\tau_n(\epsilon)$ must be smooth.

like a spectral decomposition. For $t \in (0, \infty]$, define

$$b(t) = \int_{\mathbf{X}} (1 - w/t)_+ \mathcal{Q}$$

and $b(0) = \lim_{t \downarrow 0} b(t)$. For $r > 0$ and integers $k \geq 0$, define $h_k(r) = \int_r^\infty \frac{(k+1)b(u)^k}{u^2} du$. Using a change of variables [Smith and Tierney, 1996, Theorem 1 and Lemma 3], we can write for $k \geq 0$

$$\mathcal{Q}\mathcal{P}^k(dy) = b(0)^k I_{\{w=0\}} \mathcal{Q}(dy) + h_k(w(y)) I_{\{w>0\}} \Pi(dy).$$

We defined this so that at $k = 0$, this is well defined.

Define the pushforward measures $\eta = \mathcal{Q} \circ w^{-1}$ and $\mu = \Pi \circ w^{-1}$ where w^{-1} is the inverse image of the weight function w. For integers $k \geq 0$, define $\eta_k = \mathcal{Q}\mathcal{P}^k \circ w^{-1}$. Using a change of variables like [Smith and Tierney, 1996, Theorem 1 and Lemma 3], we can get

$$\eta_k(dr) = b(0)^{k+1} \delta_0(dr) + h_k(r) \mu(dr).$$

We will use this decomposition to show that $\tau_n(\epsilon), \tau_n(1-\epsilon) < \infty$ for every fixed $n \in \mathbb{Z}_+$.

Define $d_k = \left\| \mathcal{Q}\mathcal{P}^k - \Pi \right\|_{TV}$. Define $D_k(t) = \eta_k([0,t]) - \mu([0,t])$ for $t \in [0,\infty)$. Since h_k is nonincreasing, taking the supremum, we get the identity

$$d_k = \sup_{t \geq 0} D_k(t) = \int_{(0,\infty)} (1 - h_k(r))_+ \mu(dr).$$

For $t \in [0,\infty)$, define $M(t) = \mu([0,t])$ and $M(u-) = \lim_{v \uparrow u} M(v)$. Then also by [Smith and Tierney, 1996, Equation 7], if $M(r) > 0$,

$$h_k(r) = 1 + \int_{(r,\infty)} \frac{b(u)^{k+1}}{M(u)M(u-)} \mu(du) - \frac{b(r)^{k+1}}{M(r)}.$$

So then,

$$(1 - h_k(r))_+ \leq 1 \wedge \frac{b(r)^{k+1}}{M(r)} \to 0.$$

By dominated convergence, then $d_k \to 0$ and $\tau_n(\epsilon), \tau_n(1-\epsilon) < \infty$ for every fixed $n \in \mathbb{Z}_+$.

We construct a type of multiplicative decomposition analogous to a spectral decomposition. If $M(t) = 0$, then $D_k(t) = b(0)^{k+1}$. If $M(t) > 0$, then using Fubini's theorem

$$\begin{aligned}
D_k(t) &= \mu((t,\infty)) - \eta_k((t,\infty)) \\
&= \int_{r \in (t,\infty)} \frac{b(r)^{k+1}}{M(r)} \mu(dr) - \int_{r \in (t,\infty)} \int_{u \in (r,\infty)} \frac{b(u)^{k+1}}{M(u)M(u-)} \mu(du)\mu(dr) \\
&= \int_{r \in (t,\infty)} \frac{b(r)^{k+1}}{M(r)} \mu(dr) - \int_{u \in (t,\infty)} \frac{b(u)^{k+1}[M(u-) - M(t)]}{M(u)M(u-)} \mu(du) \\
&= M(t) \int_{r \in (t,\infty)} \frac{b(r)^{k+1}}{M(r)M(r-)} \mu(dr).
\end{aligned}$$

We have proven that there is a type of multiplicative decomposition where there is a subprobability measure ν_t such that

$$D_k(t) = \int_{[0,1)} v^k \nu_t(dv)$$

and $\nu_t([0,1)) = D_0(t)$. In particular, for every Borel $A \subseteq [0,1)$, the measure is defined by

$$\nu_t(A) = \begin{cases} M(t) \displaystyle\int_{(t,\infty)} \mathbf{1}_{\{b(u)\in A\}} \frac{b(u)}{M(u)M(u-)}\, \mu(du), & M(t) > 0, \\ b(0)\, \mathbf{1}_A(b(0)), & M(t) = 0. \end{cases}$$

We use Jensen's inequality as in [Röckner and Wang, 2001, Lemma 2.2]. If $D_0(t) = 0$, then $D_k(t) = 0$ for $k \geq 0$ and so we will assume $D_0(t) > 0$. This gives for $1 \leq M \leq T$,

$$D_M(t)/D_0(t) = \int_{[0,1)} (v^T)^{M/T} \nu_t(dv)/D_0(t) \leq \left(\int_{[0,1)} v^T \nu_t(dv)/D_0(t) \right)^{M/T} \tag{9}$$

$$D_M(t) \leq D_T(t)^{M/T} D_0(t)^{1-M/T} \leq D_T(t)^{M/T} \tag{10}$$

$$d_T \geq d_M^{T/M}. \tag{11}$$

Now define for $n, k \in \mathbb{Z}_+$, $d_{n,k} = \left\| \mathcal{Q}_n \mathcal{P}_n^k - \Pi_n \right\|_{TV}$. Applying (11) and using the definition of the mixing times and the fact that the mixing times are finite, we have

$$(1-\epsilon)^{\frac{\tau_n(\epsilon)}{\tau_n(1-\epsilon)-1}} < d_{n,\tau_n(1-\epsilon)-1}^{\frac{\tau_n(\epsilon)}{\tau_n(1-\epsilon)-1}} \leq d_{n,\tau_n(\epsilon)} \leq \epsilon.$$

Taking the logarithms gives

$$\frac{\tau_n(\epsilon)}{\tau_n(1-\epsilon)} \geq \left(1 - \frac{1}{\tau_n(1-\epsilon)}\right) \frac{\log(\epsilon)}{\log(1-\epsilon)}.$$

Since $\lim_{n\to\infty} \tau_n(1-\epsilon) = \infty$, taking the limit completes the result. □

6 Unreliability of Convergence Diagnostics

The goal of this section is to further motivate the previously developed high-dimensional convergence theory. Convergence diagnostics are often used to validate convergence for Metropolis-Hastings independence samplers. The hope is that this could allow bypassing a convergence analysis entirely. We will show that a convergence diagnostic has a nontrivial probability to validate both an accurate simulation and an inaccurate sim-

ulation. In other words, we show convergence diagnostics based on the simulation output do not provide a reliable diagnostic in general.

An interesting example is the formulation of Gelman's $\hat{R}$ [Gelman and Rubin, 1992] for testing convergence in MCMC. For an index $j = 1, \ldots, J$, consider multiple IMH processes $(X_n^j(t))_{t\geq 0}$ each with Markov kernel $\mathcal{P}_n$ with target Π_n and proposal $\mathcal{Q}_n$. We use the variance-ratio statistic [Vehtari et al., 2021]. For $J \geq 2$ independent chains of length $k \geq 2$ states, let $f : \mathbf{X} \to \mathbb{R}$ be Borel measurable and write $f_{j,i} = f(X^j(i))$, and define the sample mean and sample variance $\bar{f}_j = \frac{1}{k}\sum_{i=0}^{k-1} f_{j,i}$ and $S_j^2(f) = \frac{1}{k-1}\sum_{i=0}^{k-1}(f_{j,i} - \bar{f}_j)^2$. Define the mean of the sample means $\bar{f} = \frac{1}{J}\sum_{j=1}^{J}\bar{f}_j$ and the scaled sample variance of the sample means $\overline{B}(f) = \frac{k}{J-1}\sum_{j=1}^{J}(\bar{f}_j - \bar{f})^2$ and the mean of the sample variances $\overline{S^2}(f) = \frac{1}{J}\sum_{j=1}^{J} S_j^2(f)$. The $\hat{R}(f)$ convergence diagnostic is defined by

$$\hat{R}(f) = \sqrt{\frac{(k-1)\overline{S^2}(f) + \overline{B}(f)}{k\overline{S^2}(f)}}$$

if $\overline{S^2}(f) > 0$ and $\hat{R}(f) = \infty$ if $\overline{S^2}(f) = 0$. A value $\hat{R}(f) \approx 1$ would mean the diagnostic validates convergence.

Consider an alternative sequence of IMH kernels $\mathcal{K}_n$ with target ν_n and proposal $\mathcal{Q}_n$. Let $B_n \subseteq \mathbf{X}_n$ be a sequence of Borel sets such that the weights differ only on B_n, that is, $d\Pi_n/d\mathcal{Q}_n \propto d\nu_n/d\mathcal{Q}_n$ on B_n^c and such that convergence happens

$$R_{k_n}(\mathcal{Q}_n, \nu_n) \to 0 \tag{12}$$

as $n \to \infty$.

Theorem 7. *Assume B_n is a sequence of Borel sets and $\mathcal{K}_n$ is an alternative IMH kernel with target ν_n satisfying (12). Fix $J \geq 2$ and let $k_n \geq 2$ such that $k_n \to \infty$. Assume the proposal visits the sets B_n so that*

$$\rho_n = 1 - \{1 - \mathcal{Q}_n(B_n)\}^{Jk_n} \to 0$$

as $n \to \infty$. Let $f_n : \boldsymbol{X}_n \to \mathbb{R}$ be a sequence of Borel measurable bounded functions with

$\|f_n\|_\infty \le 1$. *Assume for some* $\sigma_*^2 > 0$, *the variance is not degenerate,* $Var_{\nu_n}(f_n) \ge \sigma_*^2$ *and the means differ, that is, for some* $\Delta > 0$ *and all sufficiently large* $n \in \mathbb{Z}_+$,

$$|\Pi_n(f_n) - \nu_n(f_n)| \ge \Delta.$$

Then for every fixed $\tau > 1$ *and* $0 < \delta < \Delta$, *the* $\hat{R}(f_n)$ *diagnostic gives a false positive with high probability for the original IMH process*

$$\mathbb{P}\left(\hat{R}(f_n) \le \tau, \quad \min_{1\le j\le J}\left|\frac{1}{k_n}\sum_{i=0}^{k_n-1} f_n(X_j(i)) - \Pi_n(f_n)\right| > \delta\right) \to 1$$

as $n \to \infty$.

We first prove a more general result. Let $\mathcal{Q}$ be a proposal measure and Π the target measure and ν a comparison measure. Define the weights $w = d\Pi/d\mathcal{Q}$ and $v = d\nu/d\mathcal{Q}$. Run $J \in \mathbb{Z}_+$ independent chains, each initialized from $\mathcal{Q}$, with fixed transition counts $t_1, \ldots, t_J \in \mathbb{Z}_+$. Let $\epsilon > 0$ and let $\hat{D} \in \{0, 1\}$ be a diagnostic from the entire samples of J Markov chains of length $t_j + 1$. The number of independent raw draws from $\mathcal{Q}$, counting initializations, is

$$M = \sum_{j=1}^{J}(t_j + 1).$$

For $1 \le k \le t_j + 1$, set

$$A_{j,k}(f) = \frac{1}{k}\sum_{i=0}^{k-1} f(X_{j,i}).$$

For a specified integrable quantity $f : \mathbf{X} \to \mathbb{R}$ and tolerance $\delta > 0$,

$$\{\hat{D} = 1, |A_{1,k}(f) - \Pi(f)| > \delta\}$$

is the set of false convergence validation. Let $\mathcal{P}$ be the IMH Markov kernel with target Π and $\mathcal{K}$ with target ν. Let $\mathbb{P}_\mathcal{P}$ denote the probability with respect to $(X_t)_{t=0}^\infty$ generated from the kernel $\mathcal{P}$ and similarly define $\mathbb{P}_\mathcal{K}$ using the kernel $\mathcal{K}$.

Theorem 8. *Let* v *be a comparison weight and* w *the weight of interest, with* $w \propto v$

on B^c for some Borel set $B \subseteq \boldsymbol{X}$ with $\Pi(B) > 0$. Define $\rho = 1 - \{1 - \mathcal{Q}(B)\}^M$. Let $f : \boldsymbol{X} \to \mathbb{R}$ be Borel measurable with $\int |f| d\Pi, \int |f| d\nu < \infty$. Let $\Delta = |\Pi(f) - \nu(f)|$. For every $0 < \delta < \Delta$,

$$\begin{aligned}&\mathbb{P}_{\mathcal{P}}\left(\hat{D} = 1, \min_j |A_{j,t_j+1}(f) - \Pi(f)| \geq \Delta - \delta\right)\\ &\geq \mathbb{P}_{\mathcal{K}}(\hat{D} = 1) - \mathbb{P}_{\mathcal{K}}(\max_j |A_{j,t_j+1}(f) - \nu(f)| > \delta) - \rho.\end{aligned}$$

Proof of Theorem 8. For each j, let $Y_{j,0}, \ldots, Y_{j,t_j}$ be independent samples from $\mathcal{Q}$ and let $U_{j,k}$ be independent uniform random variables. Initialize at $Y_{j,0}$, and use the same proposal $Y_{j,i}$ and uniform for update i. Denote the coupling of $(X_{j,k})$ with target Π and $(\tilde{X}_{j,k})$ with target ν by $\mathbb{P}_2$. For $1 \leq k \leq t_j + 1$, define

$$A_{j,k}(f) = \frac{1}{k}\sum_{i=0}^{k-1} f(X_{j,i}) \qquad\qquad \tilde{A}_{j,k}(f) = \frac{1}{k}\sum_{i=0}^{k-1} f(\tilde{X}_{j,i}).$$

Let $\tilde{D}$ denote the convergence diagnostic computed only from the process $(\tilde{X}_{j,k})$ and define the sets

$$E = \{\tilde{D} = 1\} \qquad\qquad F = \left\{\max_j |\tilde{A}_{j,t_j+1}(f) - \nu(f)| \leq \delta\right\}.$$

Define the set $H = \{Y_{j,i} \notin B \text{ for all } j \text{ and } 0 \leq i \leq t_j\}$. By construction $\mathbb{P}_2(H) = \{1 - \mathcal{Q}(B)\}^M = 1 - \rho$.

On H, both chains initialize the same and are the same for all j,

$$\hat{D} = \tilde{D} \qquad\qquad A_{j,t_j+1}(f) = \tilde{A}_{j,t_j+1}(f).$$

On $E \cap F \cap H$, for all j,

$$|A_{j,t_j+1}(f) - \Pi(f)| = |\tilde{A}_{j,t_j+1}(f) - \Pi(f)| \geq |\nu(f) - \Pi(f)| - |\tilde{A}_{j,t_j+1}(f) - \nu(f)| \geq \Delta - \delta.$$

We have the lower bound

$$\begin{aligned}
&\mathbb{P}_{\mathcal{P}}\left(\hat{D}=1,\ \min_j |A_{j,t_j+1}(f)-\Pi(f)| \geq \Delta-\delta\right) \geq \mathbb{P}_2(E\cap F\cap H)\\
&\geq \mathbb{P}_2(E)-\mathbb{P}_2\left(E\cap(F^c\cup H^c)\right) \geq \mathbb{P}_2(E)-\mathbb{P}_{\mathcal{K}}(F^c)-\mathbb{P}_2(H^c)\\
&= \mathbb{P}_{\mathcal{K}}(\tilde{D}=1)-\mathbb{P}_{\mathcal{K}}\left(\max_j |\tilde{A}_{j,t_j+1}(f)-\nu(f)|>\delta\right)-\rho.
\end{aligned}$$

□

Proof of Theorem 7. Let G_n be the event that both errors are at most $\eta>0$ in every chain:

$$G_n=\bigcap_{j=1}^{J}\left\{|A_{j,k_n}(f_n)-\nu_n(f_n)|\leq\eta, |A_{j,k_n}(f_n^2)-\nu_n(f_n^2)|\leq\eta\right\}.$$

Markov's inequality gives

$$\mathbb{P}_{\mathcal{K}_n}(G_n^c)\leq\frac{2JR_{k_n}(\mathcal{Q}_n,\nu_n)}{\eta}\to 0.$$

On G_n, since $|A_{j,k_n}(f_n)|, |\nu_n(f_n)|\leq 1$,

$$\left|A_{j,k_n}(f_n)^2-\nu_n(f_n)^2\right|\leq 2|A_{j,k_n}(f_n)-\nu_n(f_n)|\leq 2\eta.$$

On G_n, then we can choose $\eta>0$ small enough depending on σ_*^2 so that

$$A_{j,k_n}(f_n^2)-(A_{j,k_n}(f_n))^2\geq\nu_n(f_n^2)-\nu_n(f_n)^2-3\eta=\mathrm{Var}_{\nu_n}(f_n)-3\eta\geq\sigma_*^2/2.$$

On G_n, $\overline{B}_n(f_n)/k_n\leq\frac{J}{J-1}\eta^2$. We can choose η in terms of τ so that we have the upper bound

$$\hat{R}(f_n)^2=1-\frac{1}{k_n}+\frac{\overline{B}_n(f_n)}{k_n}\frac{1}{\overline{S^2}_n(f_n)}\leq 1-\frac{1}{k_n}+\frac{\overline{B}_n(f_n)}{k_n}\frac{2}{\sigma_*^2}\leq\tau^2.$$

For sufficiently large $n \in \mathbb{Z}_+$, then

$$\mathbb{P}_{\mathcal{K}_n}\left(\hat{R}(f_n) \leq \tau\right) \geq \mathbb{P}_{\mathcal{K}_n}(G_n) \geq 1 - \epsilon.$$

For any $\epsilon \in (0,1)$ and $0 < \delta < \Delta$ and $j \in \{1, \ldots, J\}$, for $n \in \mathbb{Z}_+$ sufficiently large, Theorem 8 gives

$$\begin{aligned}\mathbb{P}_{\mathcal{P}_n}\left(\hat{R}(f_n) \leq \tau, |A_{j,k_n}(f_n) - \Pi_n(f_n)| \geq \Delta - \delta\right) &\geq \mathbb{P}_{\mathcal{K}_n}\left(\hat{R}(f_n) \leq \tau\right) - 2\epsilon \\ &\geq 1 - 3\epsilon.\end{aligned}$$

The conclusion follows since ϵ is arbitrary. □

7 Artificial Intelligence Usage Statement

This manuscript is written by the author in collaboration with GPT Sol 5.6 and GPT Astra through the Codex harness as a tool to develop the technical aspects of the proofs under direction from the author. In particular, the research directions and the main proof techniques are mostly contributions of the author or built heavily upon previous literature. The manuscript exposition is almost entirely contributed by the author aside from proofreading and assisting in Tikz code generation.

A Proof of Theorem 5

We first restrict the analysis to a certain set based on the assumptions on the data. Define the arithmetic mean of the Hessian of the negative log likelihood as

$$\widehat{J}_N(\beta) = \frac{1}{N}\sum_{i=1}^{N} \sigma'(x_i^T\beta) x_i x_i^T.$$

Let $X_N = (x_1, \ldots, x_N)^T$ be the design matrix with rows x_i^T, and let $c > 0$ be the constant supplied by Lemma 9. Let $\|\|_{\mathrm{op}}$ denote the operator norm. Define the set

$$E_N = \left\{ \left\| X_N^T X_N / N \right\|_{\mathrm{op}} \le 2, \|\nabla U_N(\beta_d^*)\| \le cN/4, \inf_{\|\beta\| \le B+1} \lambda_{\min}\left(\widehat{J}_N(\beta) \right) \ge c \right\}. \quad (13)$$

We will show that $\mathbb{P}_*(E_N) \to 1$.

The assumptions on the features give $\mathbb{E}_*(x_i x_i^T) = I_d$ and for every $h = (h_1, \ldots, h_d)^T$ with $\|h\| = 1$, the centered projection $x_i^T h - h_1$ has uniformly bounded sub-Gaussian norm by Vershynin [2018, Proposition 2.6.1]. A deterministic intercept h_1, with $|h_1| \le 1$, preserves this property. By Vershynin [2012, Theorem 5.39], since $d/N \to 0$, there is a constant $b > 0$ such that, for all sufficiently large $N \in \mathbb{Z}_+$,

$$\mathbb{P}_*\left(\left\| X_N^T X_N / N \right\|_{\mathrm{op}} > 2 \right) \le 2e^{-bN} \to 0.$$

By definition, the gradient

$$\nabla U_N(\beta_d^*) = \sum_{i=1}^N \left(\sigma(x_i^T \beta_d^*) - Y_i \right) x_i + \tau^{-2} \beta_d^*.$$

Using $\mathbb{E}_*\left(\|x_i\|^2 \right) = d$ and $\sigma' \le 1/4$ gives

$$\mathbb{E}_*\left(\|\nabla U_N(\beta_d^*)\|^2 \right) \le \frac{Nd}{4} + \tau^{-4} B^2.$$

For any fixed $c > 0$, Markov's inequality implies

$$\mathbb{P}_*(\|\nabla U_N(\beta_d^*)\| > cN/4) \le \frac{16}{c^2 N^2} \left(\frac{Nd}{4} + \tau^{-4} B^2 \right) \to 0.$$

The following lemma handles the more difficult set to control.

Lemma 9. *Under the assumptions on the data $(x_i, Y_i)_{i=1}^N$ and $d/N \to 0$, there is a constant $c > 0$, depending only on B and K, such that*

$$\mathbb{P}_*\left(\inf_{\|\beta\|\le B+1}\lambda_{\min}\left(\widehat{J}_N(\beta)\right)\ge c\right)\to 1.$$

Proof. Let $\beta\in\mathbb{R}^d$. The smallest eigenvalue satisfies

$$\lambda_{\min}\left(\widehat{J}_N(\beta)\right)=\inf_{\|h\|=1}h^T\widehat{J}_N(\beta)h.$$

For any $L>0$, $\sigma'(t)\ge\sigma'(L)$ for $|t|\le L$. For any $\|h\|=1$, the Cauchy-Schwarz inequality gives

$$\begin{aligned}h^T\widehat{J}_N(\beta)h=\frac{1}{N}\sum_{i=1}^N\sigma'(x_i^T\beta)(x_i^Th)^2&\ge\sigma'(L)\left[\frac{1}{N}\sum_{i=1}^N|x_i^Th|\mathbf{1}_{\{|x_i^T\beta|\le L\}}\right]^2\\&\ge\sigma'(L)\left[\frac{1}{N}\sum_{i=1}^N|x_i^Th|-\frac{1}{N}\sum_{i=1}^N|x_i^Th|\mathbf{1}_{\{|x_i^T\beta|>L\}}\right]^2.\end{aligned}$$

By Vershynin [2012, Theorem 5.39], there is a constant $b>0$ such that, for all sufficiently large N,

$$\mathbb{P}_*\left(\left\|X_N^TX_N/N\right\|_{\mathrm{op}}>2\right)\le 2e^{-bN}.$$

For every $\|\beta\|\le B+1$ and $\|h\|=1$, the Cauchy-Schwarz inequality and $\mathbf{1}_{\{|x_i^T\beta|>L\}}\le(x_i^T\beta)^2/L^2$ imply on $\{\left\|X_N^TX_N/N\right\|_{\mathrm{op}}\le 2\}$,

$$\begin{aligned}\frac{1}{N}\sum_{i=1}^N|x_i^Th|\mathbf{1}_{\{|x_i^T\beta|>L\}}&\le\left(\frac{1}{N}\sum_{i=1}^N(x_i^Th)^2\right)^{1/2}\left(\frac{1}{N}\sum_{i=1}^N\mathbf{1}_{\{|x_i^T\beta|>L\}}\right)^{1/2}\\&\le\frac{1}{L}\left(\frac{1}{N}\sum_{i=1}^N(x_i^Th)^2\right)^{1/2}\left(\frac{1}{N}\sum_{i=1}^N(x_i^T\beta)^2\right)^{1/2}\\&=\frac{\sqrt{(h^TX_N^TX_Nh)(\beta^TX_N^TX_N\beta)}}{NL}\le\frac{2(B+1)}{L}.\end{aligned}$$

By Vershynin [2018, Proposition 2.5.2], there is a constant $M>0$, depending only

on K, such that

$$\sup_{1\le i\le N}\sup_{\|h\|=1}\mathbb{E}_*\left(|x_i^T h|^4\right)\le M. \tag{14}$$

For every $\|h\|=1$, Hölder's inequality with exponents $3/2$ and 3 gives

$$\begin{aligned}\mathbb{E}_*(|x_i^T h|^2) &= \mathbb{E}_*(|x_i^T h|^{2/3}|x_i^T h|^{4/3})\\ &\le \mathbb{E}_*(|x_i^T h|)^{2/3}\mathbb{E}_*(|x_i^T h|^4)^{1/3}.\end{aligned}$$

Then

$$\mathbb{E}_*(|x_i^T h|)\ge \frac{\left\{\mathbb{E}_*\left((x_i^T h)^2\right)\right\}^{3/2}}{\left\{\mathbb{E}_*\left(|x_i^T h|^4\right)\right\}^{1/2}}\ge\frac{1}{\sqrt{M}}.$$

Taking the infimum

$$\begin{aligned}\inf_{\|h\|=1}\frac{1}{N}\sum_{i=1}^N|x_i^T h| &\ge \inf_{\|h\|=1}\frac{1}{N}\sum_{i=1}^N\mathbb{E}_*(|x_i^T h|)+\inf_{\|h\|=1}\left\{\frac{1}{N}\sum_{i=1}^N\left(|x_i^T h|-\mathbb{E}_*(|x_i^T h|)\right)\right\}\\ &\ge \frac{1}{\sqrt{M}}-\sup_{\|h\|\le 1}\left|\frac{1}{N}\sum_{i=1}^N\left(|x_i^T h|-\mathbb{E}_*(|x_i^T h|)\right)\right|.\end{aligned}$$

Let $\epsilon=(\epsilon_1,\ldots,\epsilon_N)$ be a vector of independent Rademacher random variables, independent of the data, each taking the values 1 and -1 with probability $1/2$. Write $\mathbb{E}_\epsilon$ for expectation over these random variables. Symmetrization and the Cauchy-

Schwarz inequality [Vershynin, 2018, see Exercises 6.4.4(a) and 6.7.7] yield

$$\begin{aligned}
\mathbb{E}_*\left(\sup_{\|h\|\le 1}\left|\frac{1}{N}\sum_{i=1}^N\left(|x_i^T h| - \mathbb{E}_*(|x_i^T h|)\right)\right|\right) &\le \frac{4}{N}\mathbb{E}_*\left(\mathbb{E}_\epsilon\left(\left\|\sum_{i=1}^N \epsilon_i x_i\right\|\right)\right) \\
&\le \frac{4}{N}\left(\mathbb{E}_*\left(\mathbb{E}_\epsilon\left(\left\|\sum_{i=1}^N \epsilon_i x_i\right\|^2\right)\right)\right)^{1/2} \\
&= \frac{4}{N}\left(\sum_{i=1}^N \mathbb{E}_*\left(\|x_i\|^2\right)\right)^{1/2} \\
&= 4\sqrt{\frac{d}{N}}.
\end{aligned}$$

The first equality uses $\mathbb{E}_\epsilon(\epsilon_i\epsilon_j) = 0$ for $i \neq j$ and $\mathbb{E}_\epsilon(\epsilon_i^2) = 1$; the last uses $\mathbb{E}_*(\|x_i\|^2) = d$. By Markov's inequality,

$$\begin{aligned}
\mathbb{P}_*\left(\inf_{\|h\|=1}\frac{1}{N}\sum_{i=1}^N |x_i^T h| < \frac{1}{2\sqrt{M}}\right) &\le \mathbb{P}_*\left(\sup_{\|h\|\le 1}\left|\frac{1}{N}\sum_{i=1}^N\left(|x_i^T h| - \mathbb{E}_*(|x_i^T h|)\right)\right| > \frac{1}{2\sqrt{M}}\right) \\
&\le 8\sqrt{M}\sqrt{\frac{d}{N}}.
\end{aligned}$$

Combining these, for any $\epsilon \in (0,1)$, choose N large enough so that

$$\begin{aligned}
&\mathbb{P}_*\left(\left\|X_N^T X_N/N\right\|_{\text{op}} \le 2, \inf_{\|h\|=1}\frac{1}{N}\sum_{i=1}^N |x_i^T h| \ge \frac{1}{2\sqrt{M}}\right) \\
&\quad \ge 1 - 2e^{-bN} - 8\sqrt{M}\sqrt{d/N} \ge 1 - \epsilon.
\end{aligned}$$

Choose $L = 8(B+1)\sqrt{M}$ to get with probability at least $1-\epsilon$, simultaneously for

every $\|\beta\| \le B+1$ and $\|h\| = 1$,

$$\begin{aligned}\frac{1}{N}\sum_{i=1}^{N} |x_i^T h| \mathbf{1}_{\{|x_i^T\beta| \le L\}} &= \frac{1}{N}\sum_{i=1}^{N} |x_i^T h| - \frac{1}{N}\sum_{i=1}^{N} |x_i^T h| \mathbf{1}_{\{|x_i^T\beta| > L\}} \\ &\ge \frac{1}{2\sqrt{M}} - \frac{2(B+1)}{L} = \frac{1}{4\sqrt{M}}.\end{aligned}$$

Choosing $c = \sigma'(8(B+1)\sqrt{M})/(16M) > 0$, then gives, on the same event,

$$\mathbb{P}_*\left(\inf_{\|\beta\|\le B+1}\inf_{\|h\|=1} h^T \widehat{J}_N(\beta) h \ge \sigma'(L)\left(\frac{1}{4\sqrt{M}}\right)^2 = c\right) \ge 1-\epsilon.$$

Since $\epsilon \in (0,1)$ is arbitrary, the stated conclusion follows.

□

By Lemma 9 and the previous bounds, if $d/N \to 0$,

$$\begin{aligned}\mathbb{P}_*(E_N^c) \le \mathbb{P}_*\Big(\big\|X_N^T X_N/N\big\|_{\mathrm{op}} > 2\Big) + \mathbb{P}_*(\|\nabla U_N(\beta_d^*)\| > cN/4) \\ + \mathbb{P}_*\left(\inf_{\|\beta\|\le B+1} \lambda_{\min}\Big(\widehat{J}_N(\beta)\Big) < c\right) \to 0\end{aligned}$$

and so $\mathbb{P}_*(E_N) \to 1$. We get the following consequences on E_N. For every $\|\beta\| \le B+1$,

$$\nabla^2 U_N(\beta) = N\widehat{J}_N(\beta) + \tau^{-2} I_d \succeq cN I_d.$$

For $\|\beta - \beta_d^*\| = 1/2$, integrating this Hessian bound gives

$$(\beta-\beta_d^*)^T \nabla U_N(\beta) \ge \frac{cN}{4} - \frac{1}{2}\|\nabla U_N(\beta_d^*)\| \ge \frac{cN}{8} > 0.$$

Convexity therefore places the minimizer inside this sphere, and hence

$$\left\|\widehat{\beta}_N - \beta_d^*\right\| \le \frac{1}{2}, \qquad cN I_d \preceq \widehat{H}_N \preceq CN I_d. \tag{15}$$

The upper Hessian bound follows from $\nabla^2 U_N \preceq X_N^T X_N/4 + \tau^{-2} I_d$.

We use $C > 0$ for fixed constants that may change between different inequalities. Define $s_{N,i} = \widehat{H}_N^{-1/2} x_i$ and let S_N be the matrix with rows $s_{N,i}^T$. Then the following bounds hold on E_N:

$$\begin{aligned} \left\| S_N^T S_N \right\|_{\text{op}} &\leq C, \qquad \max_{1 \leq i \leq N} \|s_{N,i}\|^2 \leq C \frac{d}{N}, \\ \sum_{i=1}^N \|s_{N,i}\|^2 &\leq Cd, \qquad \|v_N\|^2 \leq C \frac{d^2}{N}. \end{aligned} \tag{16}$$

Indeed, $S_N^T S_N = \widehat{H}_N^{-1/2} X_N^T X_N \widehat{H}_N^{-1/2}$ gives the first bound, and $\|x_i\|^2 \leq (1+K^2)d$ gives the second. Taking the trace gives the third. We have the identity

$$v_N = \frac{1}{2} \sum_{i=1}^N \sigma''(x_i^T \widehat{\beta}_N) \|s_{N,i}\|^2 s_{N,i}.$$

For the last, the definition of v_N and boundedness of σ'' give

$$\|v_N\|^2 = \frac{1}{4} \left\| S_N^T \left(\sigma''(x_i^T \widehat{\beta}_N) \|s_{N,i}\|^2 \right)_{i=1}^N \right\|^2 \leq C \sum_{i=1}^N \|s_{N,i}\|^4 \leq C \frac{d^2}{N}.$$

Define the importance weight $w_N = d\Pi_N / d\mathcal{Q}_N$. Define the Gaussian measure

$$\mathcal{G}_{\mathcal{N}} = N\left(\widehat{\beta}_N - \widehat{H}_N^{-1/2} v_N, \widehat{H}_N^{-1} \right). \tag{17}$$

Then we have the decomposition of the importance weight

$$\log(w_N) = \log\left(\frac{d\mathcal{G}_N}{d\mathcal{Q}_N} \right) + \log\left(\frac{d\Pi_N}{d\mathcal{G}_N} \right).$$

We will prove in Lemma 10 that $\|\Pi_N - \mathcal{G}_N\|_{TV} \to 0$ in $\mathbb{P}_*$ probability. For $Z_N^0 \sim N(0, I_d)$ conditional on the data, a comparison of Gaussian densities gives

$$\log\left(\frac{d\mathcal{G}_N}{d\mathcal{Q}_N} \right) \left(\widehat{\beta}_N + \widehat{H}_N^{-1/2} (Z_N^0 - v_N) \right) = \frac{\|v_N\|^2}{2} - v_N^T Z_N^0.$$

Thus, conditional on any data set with $\|v_N\| > 0$, under $\mathcal{G}_N$,

$$\frac{\log\left(d\mathcal{G}_N/d\mathcal{Q}_N\right) - \|v_N\|^2/2}{\|v_N\|} \sim N(0,1).$$

Total variation convergence implies for every such data and every $s \in \mathbb{R}$,

$$\left|\Pi_N\left(\frac{\log\left(d\mathcal{G}_N/d\mathcal{Q}_N\right) - \|v_N\|^2/2}{\|v_N\|} \leq s\right) - \Phi(s)\right| \leq \|\Pi_N - \mathcal{G}_N\|_{TV}.$$

From any subsequence, there is a further subsequence along which

$$\|\Pi_N - \mathcal{G}_N\|_{TV} \to 0, \qquad \|v_N\| \to \infty$$

hold $\mathbb{P}_*$ almost surely. For every $\epsilon > 0$,

$$\begin{aligned}(1 - e^{-\epsilon})\,\Pi_N\left(\left|\log\left(\frac{d\Pi_N}{d\mathcal{G}_N}\right)\right| > \epsilon\right) \\ \leq \int_{\mathbb{R}^d}\left|1 - \frac{d\mathcal{G}_N}{d\Pi_N}\right| d\Pi_N = 2\,\|\Pi_N - \mathcal{G}_N\|_{TV}.\end{aligned}$$

This implies $\log\left(d\Pi_N/d\mathcal{G}_N\right) \to 0$ in Π_N probability holds $\mathbb{P}_*$ almost surely.

Combining this with the transferred Gaussian limit justifies the following limit along the extracted subsequence. Therefore, $\mathbb{P}_*$ almost surely, for $s \in \mathbb{R}$,

$$\Pi_N\left(\frac{\log(w_N) - \|v_N\|^2/2}{\|v_N\|} \leq s\right) \to \Phi(s).$$

We verify the hypotheses of Lemma 4 along this further subsequence. Choose $L_N = \exp(\sqrt{\|v_N\|})$. Since $\|v_N\| \to \infty$, both $T_N(z)$ and L_N tend to infinity, and the integer part in the definition of $T_N(z)$ satisfies

$$\log\left(T_N(z)\right) - \frac{\|v_N\|^2}{2} - z\,\|v_N\| \to 0.$$

Also, $\log(L_N)/\left\|v_N\right\| = 1/\sqrt{\left\|v_N\right\|} \to 0$ and

$$\frac{\log\left(T_N(z)\right) \pm \log(L_N) - \left\|v_N\right\|^2/2}{\left\|v_N\right\|} \to z.$$

Continuity of the Gaussian distribution function Φ therefore gives

$$\Pi_N\left(\frac{w_N}{T_N(z)} \le \frac{1}{L_N}\right) \to \Phi(z), \qquad \Pi_N\left(\frac{1}{L_N} \le \frac{w_N}{T_N(z)} \le L_N\right) \to 0.$$

Lemma 4 yields

$$R_{T_N(z)}(\mathcal{Q}_N, \Pi_N) \to 2(1-\Phi(z))$$

along the almost surely convergent data subsequences. Since every subsequence has such an almost surely convergent subsequence, convergence holds in $\mathbb{P}_*$ probability. It remains to prove Lemma 10.

Lemma 10. *Under the assumptions on the data $(x_i, Y_i)_{i=1}^N$ and assuming $d^{5/3}/N \to 0$, then $\left\|\Pi_N - \mathcal{G}_N\right\|_{TV} \to 0$ in $\mathbb{P}_*$ probability, where $\mathcal{G}_N$ is defined by (17).*

Proof. As above, $C > 0$ denotes a fixed constant that may change between inequalities. The assumption implies $d^{3/2}/N \le d^{5/3}/N \to 0$, and hence also $d/N \to 0$. Define u_N by

$$u_N(z) = U_N(\widehat{\beta}_N + \widehat{H}_N^{-1/2} z) - U_N(\widehat{\beta}_N).$$

We have $u_N(0) = 0$ and $\nabla u_N(0) = 0$. Let P_N and G_N be the images of Π_N and $\mathcal{G}_N$ under the invertible affine change of variables $\beta \mapsto \widehat{H}_N^{1/2}(\beta - \widehat{\beta}_N)$. For every Borel set $A \subseteq \mathbb{R}^d$,

$$P_N(A) = \Pi_N(\widehat{\beta}_N + \widehat{H}_N^{-1/2} A), \qquad G_N(A) = \mathcal{G}_N(\widehat{\beta}_N + \widehat{H}_N^{-1/2} A).$$

Then $P_N(dz) \propto e^{-u_N(z)}\, dz$ and $G_N = N(-v_N, I_d)$. Choose $c > 0$ to be the constant in Lemma 9. Choose a fixed $0 < \rho \le \sqrt{c}/4$, set $r_N = \rho\sqrt{N}$, and write $B_{r_N} = \{z : \|z\| \le$

$r_N\}$. Let $Z_N \sim G_N$ and define

$$\eta_N^2 = \mathbb{E}_{G_N}\left(\|\nabla u_N(Z_N) - Z_N - v_N\|^2\right).$$

On E_N, (15) gives $\left\|\widehat{H}_N^{-1/2} z\right\| \le 1/2$ whenever $\|z\| \le 2r_N$. For $\|z\| \le 2r_N$,

$$\left\|\widehat{\beta}_N + \widehat{H}_N^{-1/2} z\right\| \le \|\beta_d^*\| + \left\|\widehat{\beta}_N - \beta_d^*\right\| + \left\|\widehat{H}_N^{-1/2} z\right\| \le B + \frac{1}{2} + \frac{1}{2} = B + 1.$$

For $\|z\| \le 2r_N$, $\nabla^2 U_N \succeq cNI_d$ and $\widehat{H}_N \preceq CNI_d$ and let $m = \frac{c}{C}$, which is a fixed constant. Then

$$\nabla^2 u_N(z) = \widehat{H}_N^{-1/2} \nabla^2 U_N\left(\widehat{\beta}_N + \widehat{H}_N^{-1/2} z\right) \widehat{H}_N^{-1/2} \succeq cN\widehat{H}_N^{-1} \succeq mI_d.$$

Since $\nabla^2 u_N \succeq mI_d$ on B_{r_N} for a fixed $m > 0$ independent of N, d, and the data, then Katsevich [2025, Lemma 2.11], applied with potentials $u_N(z)$ and $\|z + v_N\|^2/2$, yields the upper bound

$$\|\Pi_N - \mathcal{G}_N\|_{TV} = \|P_N - G_N\|_{TV} \le P_N(B_{r_N}^c) + G_N(B_{r_N}^c) + \frac{\eta_N}{2\sqrt{mG_N(B_{r_N})}}. \tag{18}$$

On E_N, we have the upper bound $\|v_N\|^2 \le Cd^2/N$ in (16) and using Markov's inequality gives

$$G_N(B_{r_N}^c) \le \frac{d + \|v_N\|^2}{r_N^2} \le \frac{C}{\rho^2}\left(\frac{d}{N} + \frac{d^2}{N^2}\right) \to 0.$$

For each fixed data set, the Gaussian prior ensures Gaussian decay of $e^{-u_N(z)}$, and $\nabla u_N(z)$ grows at most linearly. Integration by parts therefore gives

$$0 = \int_{\mathbb{R}^d} \operatorname{div}\left(ze^{-u_N(z)}\right) dz = \int_{\mathbb{R}^d} \left(d - z^T \nabla u_N(z)\right) e^{-u_N(z)}\, dz,$$

and hence

$$\int_{\mathbb{R}^d} z^T \nabla u_N(z)\, P_N(dz) = d.$$

Convexity and $\nabla u_N(0) = 0$ imply $z^T \nabla u_N(z) \ge 0$ for every $z \in \mathbb{R}^d$, which we use with

Markov's inequality. On E_N, local curvature at radius r_N and monotonicity of radial derivatives give $z^T \nabla u_N(z) \geq m r_N^2$ for $\|z\| \geq r_N$. Thus

$$P_N(B^c_{r_N}) \leq \frac{1}{m r_N^2} \int_{\mathbb{R}^d} z^T \nabla u_N(z)\, P_N(dz) = \frac{d}{m\rho^2 N} \to 0.$$

It remains to prove on E_N that $\eta_N^2 \to 0$. Taking the derivative and using $\nabla U_N(\widehat{\beta}_N) = 0$ gives, for every $z \in \mathbb{R}^d$,

$$\begin{aligned}
\nabla u_N(z) &= \widehat{H}_N^{-1/2}\big[\nabla U_N(\widehat{\beta}_N + \widehat{H}_N^{-1/2} z) - \nabla U_N(\widehat{\beta}_N)\big] \\
&= \sum_{i=1}^N \big[\sigma(x_i^T \widehat{\beta}_N + s_{N,i}^T z) - \sigma(x_i^T \widehat{\beta}_N)\big] s_{N,i} + \tau^{-2} \widehat{H}_N^{-1} z \\
&= \sum_{i=1}^N \sigma'(x_i^T \widehat{\beta}_N)(s_{N,i}^T z) s_{N,i} + \frac{1}{2} \sum_{i=1}^N \sigma''(x_i^T \widehat{\beta}_N)(s_{N,i}^T z)^2 s_{N,i} + \tau^{-2} \widehat{H}_N^{-1} z + r_{N,3}(z)
\end{aligned}$$

where the remainder

$$r_{N,3}(z) = \frac{1}{2} \sum_{i=1}^N (s_{N,i}^T z)^3 s_{N,i} \int_0^1 (1-t)^2 \sigma'''(x_i^T \widehat{\beta}_N + t s_{N,i}^T z)\, dt.$$

On E_N,

$$\|r_{N,3}(z)\|^2 \leq \frac{\|\sigma'''\|_\infty^2}{36} \big\|S_N^T S_N\big\|_{\mathrm{op}} \sum_{i=1}^N |s_{N,i}^T z|^6 \leq C \sum_{i=1}^N |s_{N,i}^T z|^6.$$

We also have

$$\begin{aligned}
\sum_{i=1}^N \sigma'(x_i^T \widehat{\beta}_N)(s_{N,i}^T z) s_{N,i} &= \widehat{H}_N^{-1/2} \left[\sum_{i=1}^N \sigma'(x_i^T \widehat{\beta}_N) x_i x_i^T\right] \widehat{H}_N^{-1/2} z \\
&= \widehat{H}_N^{-1/2} \left(\widehat{H}_N - \tau^{-2} I_d\right) \widehat{H}_N^{-1/2} z \\
&= z - \tau^{-2} \widehat{H}_N^{-1} z.
\end{aligned}$$

Thus, for every $z \in \mathbb{R}^d$,

$$\nabla u_N(z) - z = \frac{1}{2}\sum_{i=1}^{N} \sigma''(x_i^T \widehat{\beta}_N)(s_{N,i}^T z)^2 s_{N,i} + r_{N,3}(z),$$
$$\|r_{N,3}(z)\|^2 \leq C \sum_{i=1}^{N} |s_{N,i}^T z|^6. \tag{19}$$

Write $Z_N = Z_N^0 - v_N$ with $Z_N^0 \sim N(0, I_d)$. The expectations involving Z_N^0 below are conditional on the observed data, so $s_{N,i}$ and v_N are constants here. Then expanding the square

$$\begin{aligned}\nabla u_N(Z_N) - Z_N - v_N = \frac{1}{2}\sum_{i=1}^{N} \sigma''(x_i^T \widehat{\beta}_N) s_{N,i} \left[(s_{N,i}^T Z_N^0)^2 - \|s_{N,i}\|^2\right] \\ - \sum_{i=1}^{N} \sigma''(x_i^T \widehat{\beta}_N) s_{N,i}(s_{N,i}^T v_N)(s_{N,i}^T Z_N^0) \\ + \frac{1}{2}\sum_{i=1}^{N} \sigma''(x_i^T \widehat{\beta}_N) s_{N,i}(s_{N,i}^T v_N)^2 + r_{N,3}(Z_N).\end{aligned}$$

For the Taylor remainder, (19) gives

$$\begin{aligned}\mathbb{E}\left(\|r_{N,3}(Z_N)\|^2\right) &\leq C \sum_{i=1}^{N} \mathbb{E}\left(|s_{N,i}^T Z_N^0 - s_{N,i}^T v_N|^6\right) \\ &\leq C \sum_{i=1}^{N} \left[\mathbb{E}\left(|s_{N,i}^T Z_N^0|^6\right) + |s_{N,i}^T v_N|^6\right] \\ &\leq C \sum_{i=1}^{N} \left[\|s_{N,i}\|^6 + |s_{N,i}^T v_N|^6\right].\end{aligned}$$

By (16),

$$\sum_i \left[(s_{N,i}^T v_N)^2 \|s_{N,i}\|^2 + \|s_{N,i}\|^6\right] \le C\frac{d^3}{N^2},$$

$$\sum_i (s_{N,i}^T v_N)^4 \le C\frac{d}{N}\|v_N\|^4.$$

Moreover, $\max_{1\le i\le N} |s_{N,i}^T v_N|^2 \le Cd^3/N^2 \to 0$, since $d^3/N^2 = (d^{3/2}/N)^2 \to 0$. Thus, for all sufficiently large N, uniformly on E_N,

$$\sum_{i=1}^N |s_{N,i}^T v_N|^6 \le \left(\max_{1\le i\le N} |s_{N,i}^T v_N|^2\right) \sum_{i=1}^N (s_{N,i}^T v_N)^4 \le \sum_{i=1}^N (s_{N,i}^T v_N)^4.$$

Using the boundedness of σ'' and the bounds (16),

$$\begin{aligned}
&\mathbb{E}\left(\left\|\sum_{i=1}^N \sigma''(x_i^T\widehat{\beta}_N) s_{N,i}(s_{N,i}^T v_N)(s_{N,i}^T Z_N^0)\right\|^2\right) \\
&\quad \le C\sum_{i=1}^N (s_{N,i}^T v_N)^2 \mathbb{E}\left((s_{N,i}^T Z_N^0)^2\right) \\
&\quad = C\sum_{i=1}^N (s_{N,i}^T v_N)^2 \|s_{N,i}\|^2.
\end{aligned}$$

The same bounds control the constant term in the decomposition

$$\left\|\frac{1}{2}\sum_{i=1}^N \sigma''(x_i^T\widehat{\beta}_N) s_{N,i}(s_{N,i}^T v_N)^2\right\|^2 \le C\sum_{i=1}^N (s_{N,i}^T v_N)^4.$$

For each i, j,

$$\begin{aligned}
&\mathbb{E}\left(\left[(s_{N,i}^T Z_N^0)^2 - \|s_{N,i}\|^2\right]\left[(s_{N,j}^T Z_N^0)^2 - \|s_{N,j}\|^2\right]\right) \\
&= \mathbb{E}\left((s_{N,i}^T Z_N^0)^2 (s_{N,j}^T Z_N^0)^2\right) - \|s_{N,i}\|^2 \|s_{N,j}\|^2 \\
&= 2(s_{N,i}^T s_{N,j})^2.
\end{aligned}$$

This implies the identity

$$\begin{aligned}
&\mathbb{E}\left(\left\|\frac{1}{2}\sum_{i=1}^{N}\sigma''(x_i^T\widehat{\beta}_N)s_{N,i}\left[(s_{N,i}^TZ_N^0)^2-\|s_{N,i}\|^2\right]\right\|^2\right)\\
&\quad=\frac{1}{4}\sum_{i,j=1}^{N}\sigma''(x_i^T\widehat{\beta}_N)\sigma''(x_j^T\widehat{\beta}_N)(s_{N,i}^Ts_{N,j})\\
&\qquad\times\mathbb{E}\left(\left[(s_{N,i}^TZ_N^0)^2-\|s_{N,i}\|^2\right]\left[(s_{N,j}^TZ_N^0)^2-\|s_{N,j}\|^2\right]\right)\\
&\quad=\frac{1}{2}\sum_{i,j=1}^{N}\sigma''(x_i^T\widehat{\beta}_N)\sigma''(x_j^T\widehat{\beta}_N)(s_{N,i}^Ts_{N,j})^3.
\end{aligned}$$

Since $\widehat{H}_N \succeq cNI_d$, $0 \preceq S_NS_N^T \preceq (cN)^{-1}X_NX_N^T$, then the Schur product theorem implies

$$\left[(s_{N,i}^Ts_{N,j})^3\right]_{i,j=1}^N \preceq \frac{1}{(cN)^3}\left[(x_i^Tx_j)^3\right]_{i,j=1}^N.$$

Combining the preceding estimates gives, on E_N and for all sufficiently large N,

$$\eta_N^2 \le C\left(\frac{1}{N^3}\sum_{i,j=1}^{N}|x_i^Tx_j|^3+\frac{d^3}{N^2}+\frac{d}{N}\|v_N\|^4\right). \tag{20}$$

For $i \neq j$, row independence allows us to use (14) conditionally on x_i. Hölder's inequality with exponents $4/3$ and 4 then yields

$$\begin{aligned}
\mathbb{E}_*\left(|x_i^Tx_j|^3 \mid x_i\right) &\le \left\{\mathbb{E}_*\left(|x_i^Tx_j|^4 \mid x_i\right)\right\}^{3/4}\\
&\le M^{3/4}\|x_i\|^3\\
&\le Cd^{3/2}.
\end{aligned}$$

For the diagonal $i = j$, $|x_i^T x_i|^3 = \|x_i\|^6 \leq (1 + K^2)^3 d^3$. These imply the upper bound

$$\mathbb{E}_*\left(\frac{1}{N^3}\sum_{i,j=1}^{N}|x_i^T x_j|^3\right) \leq \frac{C}{N^3}\left(N(N-1)d^{3/2} + Nd^3\right)$$
$$\leq C\left(\frac{d^{3/2}}{N} + \frac{d^3}{N^2}\right) \to 0.$$

By Markov's inequality, the first term in the bound for η_N^2 tends to zero in $\mathbb{P}_*$ probability. The second term tends to zero since $d^3/N^2 = (d^{3/2}/N)^2 \to 0$.

For the third term, (16) gives, uniformly on E_N,

$$\frac{d}{N}\|v_N\|^4 \leq C\frac{d^5}{N^3} = C\left(\frac{d^{5/3}}{N}\right)^3 \to 0.$$

Since $\mathbb{P}_*(E_N) \to 1$, it follows that $\eta_N^2 \to 0$ in $\mathbb{P}_*$ probability. The two tail bounds also tend to zero uniformly on E_N, so $G_N(B_{r_N}) \to 1$ in $\mathbb{P}_*$ probability. Substituting these limits and $\eta_N^2 \to 0$ into (18) proves $\|\Pi_N - \mathcal{G}_N\|_{TV} \to 0$ in $\mathbb{P}_*$ probability.

□

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